

Executive Summary

We analyzed customer behavior and found that response rates vary dramatically across four customer groups, from 3.5% to 25.2%. These differences are real and not explained by income alone. By focusing marketing on Clusters 1, 2, and 4, we can reach 76% of customers while capturing 94% of all expected responses, which means we avoid spending money on customers who rarely respond. A deeper review showed that the higher response rates come mainly from customer behavior not from the cluster label itself. Especially how recently they purchased and how active they are. However this is good news as the clusters are still extremely useful because they group together the customers who are already most engaged. Our recommendation is to continue targeting Clusters 1, 2, and 4, and avoid Cluster 3. This approach saves budget, improves ROI, and keeps marketing focused on the customers most likely to respond.

More detailed version for expert consultant

We performed clustering, bootstrap inference, income-stratified ATE estimation, permutation testing, and a regression-adjusted causal analysis using the Lin estimator. The raw response gap between high-value clusters and the remaining customers was approximately fifteen percentage points, and this difference remained around fifteen percent even after controlling for income, which shows that income alone does not explain the gap. After adjusting for additional confounders such as age, education, marital status, number of children, recency, and purchase-channel activity, the regression-adjusted ATE decreased to 2.97 percent with a 95 percent confidence interval ranging from -4.23 to 10.16 and a p-value of 0.4188. This indicates that most of the observed difference in response rates is driven by underlying customer behavior rather than the cluster assignment itself. The clusters remain operationally useful because they group together customers who are naturally more engaged and therefore more likely to respond, even though the cluster label is not an independent causal driver of response. The overall conclusion is that the targeting strategy remains valid, but the reason it works is behavioral rather than causal, and selective advertising continues to be the most effective business decision.